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Advanced Computer Architecture

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Name
Last Name
Code Name (Codice Persona)

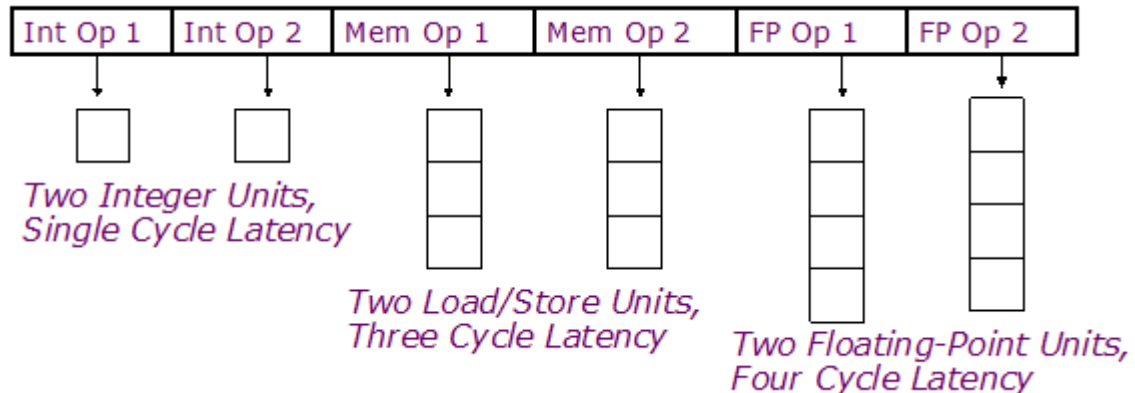
WRITE YOUR NAME and CODE NAME.

Pick 2 out of th 3 problems, **DO NOT DO ALL THE 3!**

Questions (from Forms) (20%)	
Problem 1 (40%)	
Problem 2 (40%)	
Problem 3 (40%)	
Total (100%)	

Problem 1

Consider the following VLIW “architecture” with multi-cycled unpipelined functional units.



Considering the following portion of assembly.

```
LOOP: beq $t6,$t7, END
      lw $t2,VECTB($t6)
      lw $t3,VECTC($t6)
      sw $t2,VECTA($t6)
      addi $t3,$t3,4
      sw $0,VECTD($t6)
      sw $t3,VECTC($t6)
      addi $t6,$t6,4
      blt $t6,$t7, LOOP
```

1.A schedule the following code for the VLIW with an IN-ORDER ISSUE. Branch completed with 1 cycle delay slot (branch solved in ID stage).

1.B How the does the scheduling change if we consider an IN-ORDER ISSUE and pipelined FU?

Answer 1.A

Postponing by two CC for control hazard has been considered correct as well

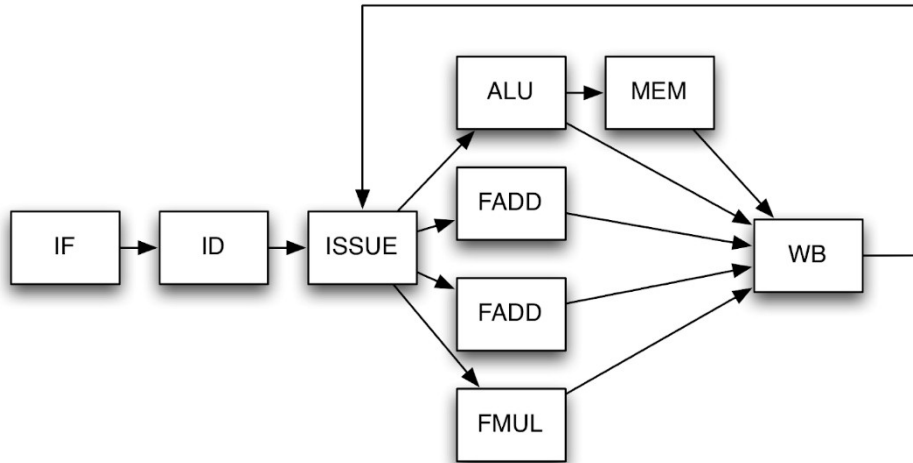
	INT1	INT2	MU1	MU2	FPU1	FPU2
C1	FOR:beq \$t6,\$t7, END		lw \$t2,VECTB(\$t6)	lw \$t3,VECTC(\$t6)		
C2						
C3						
C4	addi \$t3,\$t3,4		sw \$t2,VECTA(\$t6)	sw \$t0,VECTD(\$t6)		
C5						
C6						
C7	addi \$t6,\$t6,4		sw \$t3,VECTC(\$t6)			
C8						
C9	blt \$t6,\$t7, FOR					
C10						
C11						
C12						
C13						
C14						
C15						

Answer 1.B

	FU1	FU2	FU3	FU4	FU5	Notes
C1	FOR:beq \$t6,\$t7, END		lw \$t2,VECTB(\$t6)	lw \$t3,VECTC(\$t6)		
C2						
C3						
C4	addi \$t3,\$t3,4		sw \$t2,VECTA(\$t6)	sw \$t0,VECTD(\$t6)		
C5	addi \$t6,\$t6,4		sw \$t3,VECTC(\$t6)			
C6						
C7	blt \$t6,\$t7, FOR					
C8						
C9						
C10						
C11						
C12						
C13						
C14						
C15						

Problem 2

In this problem we will examine the execution of a code segment on the following single-issue out-of-order processor:



You can assume that:

- All functional units are pipelined
- ALU operations take 1 cycle
- MEM operations take 3 cycles (includes time in ALU)
- FP add instructions take 3 cycles
- FP multiply instructions take 5 cycles
- The WB unit has a single write port.
- RF can write first half of the CC and Read in the second half.
- There is no register renaming
- Instructions are fetched, decoded and issued in order
- The issue stage is a buffer of unlimited length that holds instructions waiting to start execution
- An instruction will only enter the issue stage if it does not cause a WAR or WAW hazard.
- Only one instruction can be issued at a time, and in the case multiple instructions are ready, the oldest one will go first

LD	F1, 0(R1)
LD	F2, 0(R1)
ADDD	F1, F1, F2
ADDI	R1, R1, 8
MULD	F1, F1, F3
SD	F1, 16(R1)

1.A List all the possible dependencies/conflicts in the code

1.B Solve considering: WAW and WAR are solved in the FIRST HALF of the WB.

1.C Consider now that the WARs are solved at the END of the ISSUE.

Answer 1.A

RAW I1-I3 F1
 RAW I2-I3 F2
 RAW I3-I5 F1
 RAW I4-I6 R1
 RAW I5-I6 F1

WAW I1-I3 F1
 WAW I3-I5 F1
 WAW I1-I5 F1

WAR I3-I5 F1
 WAR I1-I4 R1
 WAR I2-I4 R1

Answer 1.B

Both execution are acceptable. So view where the stage completes in correspondence of the symbol or viceversa where the stage begins

[illegible][illegible]

Answer 1.C

Is the same execution table as before

[illegible]

Problem 3

Assume that the following code is executed on a CPU with Tomasulo

Instruction	ISSUE	START EXE	WB
I1: LD F6 32+ R2	1	2	5
I2: LD F2 45+ R3	2	6	9
I3: MULTD F0 F4 F3	3	4	15
I4: ADD F8 F2 F6	4	10	16
I5: DIVD F12 F8 F0	5	17	28
I6: SUBD F8 F6 F2	17	18	23

A. List all the possible dependencies/conflicts in the code.

B. Is there a “configuration” that can respect the shown execution?
How many units? Which kind? What latency? How many Reservation Stations?

C. If the previous table was not correct, please, write the right one.

Answer 3.A

RAW F6 I1-I4
RAW F6 I1-I6

RAW F2 I2-I4
RAW F2 I2-I6

RAW F0 I3-I5
RAW F8 I4-I5

WAR F8 I5-I6
WAW F8 I4-I6

Answer 3.B

The table is correct.

One of the configurations that can respect the shown table is

RS1, RS2 + 1 LDU (3cc)
RS3, RS4 + 1 MUL (11cc)
RS5 + 1 ADD (5cc)

Consider structural hazards for RS, FU and CDB.

Answer 3.C

Instruction	ISSUE	START EXE	WB
I1: LD F6 32+ R2			
I2: LD F2 45+ R3			
I3: MULTD F0 F4 F3			
I4: ADD F8 F2 F6			
I5: DIVD F12 F8 F0			
I6: SUBD F8 F6 F2			